In-Bum Chung

Research focus: **Data-driven design** for **complex engineering systems**; learning **representations** of continuous, categorical, and structural design domain; advancing optimization with **surrogate modeling** and **adaptive sampling**; **system-level design** for infrastructure networks and manufacturing systems

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Education

University of Illinois Urbana-Champaign — **Illinois, United States**
Ph.D. in Industrial Engineering — *August 2026*
Dissertation: **Design and recovery optimization of complex network systems for resilience enhancement**
Advisor: Dr. Pingfeng Wang

Hanyang University — **Seoul, Republic of Korea**
M.S. in Mechanical Engineering — *February 2018*
Thesis: **Dimensionality reduction using statistical analysis and model based methods: a comparison between elementary effect method and random forest regressor**
Advisor: Dr. Dong-Hoon Choi

Hanyang University — **Seoul, Republic of Korea**
B.S. in Mechanical Engineering (Cum Laude) — *February 2016*

Work Experience

Northwestern University (Mechanical Engineering) — **Illinois, United States**
Postdoctoral Scholar (Advisor: Dr. Wei Chen) — *August 2026-Current*
- Research projects on autonomous materials discovery

University of Illinois Urbana-Champaign (Industrial & Systems Engineering) — **Illinois, United States**
Graduate Research Assistant (Advisor: Dr. Pingfeng Wang) — *August 2022-July 2026*
- Research projects on hybrid energy systems and remanufacturing

PIDOTECH Inc. — **Seoul, Republic of Korea**
Research Engineer (Full-time) — *February 2018-June 2021*
Research Engineer (Part-time) — *July 2021-June 2022*
- R&D of optimization algorithms, metamodeling methods, and dimensionality reduction techniques
- Investigation and implementation of deep-learning models and methods for engineering design applications

Research Projects

- **HT-MAX: High-throughput Materials Discovery for Extremes** (10/2026 – Ongoing)
  - Project funded by U.S. Army Research Laboratory: creating a machine learning model to find the relationship between the materials composition, structure, and properties of compositionally complex ceramics; build ML and data-driven tools for quantifying uncertainties and guide adaptive discovery (Bayesian optimization)

- **Accelerated Design, Discovery, and Deployment of Electronic Phase Transitions** (09/2026 – Ongoing)
  - Project funded by NSF DMREF: design and discovery materials exhibiting insulator-to-metal transitions for semiconductors using AI-aided screening and adaptive discovery; materials design optimization using Bayesian optimization and LVGP - a Gaussian process model that can handle categorical variables

- **Data-Driven Design Decision Support for Remanufacturing of High-Value Components in Industrial and Agricultural Equipment** (11/2023 – 09/2025)
  - Project funded by REMADE Institute: collaborating with industrial partner (John Deere) to create models for

evaluating economic and environmental impact of manufacturing and remanufacturing equipment, conducting failure mode analysis, cost analysis, and establish framework for design for remanufacturing (DfRem); expand concept to system-level with multiple parts; estimate system behavior through data-driven modeling

- **Multi-timescale Nuclear-Renewable Hybrid Energy Systems Operations to Improve Electricity System Resilience, Reliability, and Economic Efficiency** (09/2022 – 12/2023)
  - Project funded by DOE: establishing an open-source repository to share the dataset and codes for resilient power networks through deep generative approach for disruption management; establishing a lab setup for hardware-in-the-loop (HIL) test for integrated energy system (IES) control

Journal Publications *(*co-first authorship)*

[J1] **In-Bum Chung**, Pingfeng Wang, "ROCIN: Recovery Optimization of Coupled Infrastructure Networks for Resilient Dispatch on Dynamic Power-Transportation Network," Reliability Engineering & System Safety, 2026 *(Submitted*).

[J2] **In-Bum Chung**, Yi Luo, Pingfeng Wang, "Data-driven Co-design of Power Distribution Networks for Resilience Enhancement through Graph Neural Network aided Performance Estimation," Reliability Engineering & System Safety, 112511, 2026.

[J3] Mohammad Mundiwala, **In-Bum Chung** … Pingfeng Wang, Chao Hu, "Probabilistic System-Level Cost Modeling for Design for Remanufacturing," Journal of Mechanical Design, 148(9): 092001, 2026.

[J4] **In-Bum Chung**, Pingfeng Wang, "Design of Transmission Networks for Enhanced Resilience under Stochastic Disruption Scenarios using Graph Generative Models," Journal of Mechanical Design, 148(2): 021706, 2026.

[J5] **In-Bum Chung**, Pingfeng Wang, "Dataset on Complex Power Systems: Design for Resilient Transmission Networks using a Generative Model," Journal of Mechanical Design, 147(4): 041709, 2025.

[J6] Xinyang Liu*, **In-Bum Chung***, Mohammad Behtash*, Matthew Davied, Todd Thompson … Pingfeng Wang, Chao Hu, "A Design for Remanufacturing Framework Incorporating Identification, Evaluation, and Validation: A Case Study of Hydraulic Manifold," Journal of Mechanical Design, 147(8): 084502, 2025.

[J7] Jiaxin Wu, **In-Bum Chung**, Zheng Liu, Pingfeng Wang, "Co-Design Optimization of Combined Heat and Power-based Microgrids," *Journal of Renewable and Sustainable Energy*, 15(5): 056301, 2023.

[J8] **In-Bum Chung**, Dohyun Park, Dong-Hoon Choi, "Surrogate-based Global Optimization Using an Adaptive Switching Infill Sampling Criterion for Expensive Black-Box Functions," *Structural and Multidisciplinary Optimization*, 57, 1443-1459, 2018.

[J9] Yong-Hun Kang, **In-Bum Chung**, Dong-Hoon Choi, "Simulation-based Turbofan Shape Optimization for Reducing Power Consumption and Noise of a Bladeless Circular Ceiling Air Conditioner," *International Journal of Precision Engineering and Manufacturing*, 18, 1155-1163, 2017.

Conference Proceedings

[C1] **In-Bum Chung**, Pingfeng Wang, "Infrastructure network Expansion Using a Conditional Graph Generative Model for Sequentially Estimating Connections to Maximize System Resilience," In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, 2026 (*Presented*).

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[C2] **In-Bum Chung**, Lei Wu, Pingfeng Wang. "Recovery Optimization of a Power Distribution Network Considering a Coupling with Transportation Network for Dispatching Resources." In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol. 89220, p. V03AT03A045. American Society of Mechanical Engineers, 2025.

[C3] Mohammad Mundiwala, **In-Bum Chung**, Matthew Davied, Michael Lee, Corey Smith… Chao Hu. "A System-Level Cost Modeling Framework for Design for Remanufacturing: a Case Study of an Agricultural Machine Transmission." In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol. 89237, p. V03BT03A038. American Society of Mechanical Engineers, 2025.

[C4] Parth Bansal, **In-Bum Chung**, Mohammad Mundiwala, Chao Hu, Pingfeng Wang. "Fault Detection and Pressure-Time Curve Prediction for Fluid-Structure Interactions with Physics-Based Modeling and Machine Learning." In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol. 89220, p. V03AT03A044. American Society of Mechanical Engineers, 2025.

[C5] **In-Bum Chung**, Yi Luo, Pingfeng Wang, "Disruption Management of Interdependent Power Networks Using a Data-Driven Co-Design Approach for Enhanced System Resilience." In ASME *International Mechanical Engineering Congress and Exposition*, vol. 88599, p. V001T02A015. American Society of Mechanical Engineers, 2024.

[C6] Xinyang Liu, **In-Bum Chung**, Mohammad Behtash, Matthew Davied, Todd Thompson… Chao Hu. "A Design for Remanufacturing Framework Incorporating Identification, Evaluation, and Validation: A Case Study of Hydraulic Manifold." In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol. 88377, p. V03BT03A049. American Society of Mechanical Engineers, 2024.

[C7] **In-Bum Chung**, Pingfeng Wang, "Generative Design for Power System Networks Using WGAN and Graph Performance Measures for Guided Generation," In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol. 88360, p. V03AT03A040. American Society of Mechanical Engineers, 2024.

[C8] Zheng Liu, Jiaxin Wu, Wuchen Fu, Pouya Kabirzadeh, **In-Bum Chung**… Yumeng Li. "Control Co-Design of Battery Packs with Immersion Cooling." In *ASME International Mechanical Engineering Congress and Exposition*, vol. 87592, p. V002T02A016. American Society of Mechanical Engineers, 2023.

[C9] **In-Bum Chung**, Pingfeng Wang, "Multi-Fidelity Modeling for Dynamic Power Control and Optimization of Nuclear-Renewable Hybrid Energy Systems." In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol. 87301, p. V03AT03A036. American Society of Mechanical Engineers, 2023.

[C10] **In-Bum Chung**, Jiaxin Wu, and Pingfeng Wang. "Control Co-Design of Combined Heat and Power Hybrid Energy Systems." In *IISE Annual Conference and Expo*. IISE, 2023.

[C11] Dohyun Park, **In-Bum Chung**, Dong-Hoon Choi. "Surrogate based Global Optimization Using Adaptive Switching Infill Sampling Criterion." In *World Congress of Structural and Multidisciplinary Optimisation*, pp. 692-699. Cham: Springer International Publishing, 2017.

Awards & Fellowships

Jerry S. Dobrovolny Graduate Student Fellowship (UIUC) AY 2025-2026
Mavis Future Faculty Fellow (UIUC)* AY 2025-2026
**MF3 program trains fellows on the responsibilities of a faculty member - research, teaching, and service*
Ben Hamilton Graduate Research Award (UIUC) 2024

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Hansen Fellowship (UIUC) 2022

Teaching Experience

Certificate in Foundations of Teaching (UIUC CITL) – involves teaching development workshops, observation and reflection of teaching, and exploring literature on teaching

Teaching Assistantship – UIUC (**Recognized on the "**List of Teachers Ranked as Excellent by Their Students**")*
SE 101: Engineering Graphics & Design* (Undergraduate Core) Spring 2025
- Teaching lab sessions, exam proctor, organizing office hours, and grading assignments & exams

SE 410: Components Design (Undergraduate/Graduate Elective) Fall 2024
SE 450: Decision Analysis I (Graduate Elective) Fall 2023
- Organizing office hours, exam review sessions, and grading assignments & exams

Teaching Assistantship – Hanyang University
APA 4010: Mechanism Design (Undergraduate Elective) Fall 2017 & Fall 2016
DME 3052: Machine Elements Design (Undergraduate Elective) Spring 2017
- Organizing office hours, exam review sessions, and grading assignments & exams

Professional Service

Served as a reviewer for conferences: IDETC2025
Served as a reviewer for journal paper: The Journal of Supercomputing

Skills

Programming: **Python** (Numpy, Scipy, Scikit-Learn, PyTorch, TensorFlow), MATLAB, C
Modeling: **Gaussian Process** (DACE, GPyTorch)**, Network/Graph-based modeling** (NetworkX), power system modeling (MATPOWER, pandapower), **machine learning-based surrogate modeling**, design of experiments
Optimization: **Data-driven optimization, Bayesian Optimization, metaheuristic optimization**, stochastic optimization, mixed-integer programming
CAD Software: CATIA, Autodesk Fusion 360

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